Q.28 Define fluctuation of speed and fluctuation of energy in a flywheel.

Q.29 Draw and explain uniform velocity cam displacement diagram.

Q.30 Explain the different types of cams.

Q.31 Explain balancing of a single rotating mass situated in one plane.

Q.32 Describe the steps for balancing several masses rotating in different planes.

Q.33 Explain transverse vibration in shafts.

Q.34 List harmful effects of vibrations in machines.

Q.35 Write short note on chain drives and terminology.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 With neat sketches, explain different inversions of double slider crank chain.

Q.37 Derive the expression for ratio of tensions in a flat belt drive and condition for maximum power.

Q.38 The torque in an I.C. engine varies between 400 N.m (max) and 150 N.m (min) during a revolution. The mean speed is 280 rpm, and the coefficient of fluctuation of speed is not to exceed 0.02. The flywheel has a radius of gyration of 0.65 m.

Find :

a) Fluctuation of energy per revolution

b) Mass of the flywheel required.

(20)

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6th Sem / Mecatronics
Subject:- Mechanisms and Machines

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A kinematic pair with surface contact is called:

a) Lower pair b) higher pair

c) Turning pair d) Sliding pair

Q.2 The number of kinematic pairs in a four-bar chain is:

a) 2 b) 4

c) 3 d) 6

Q.3 An open belt drive runs between two pulleys of diameters 500 mm and 250 mm with center distance 2 m. The belt used is :

a) Open b) Cross

c) Compound d) None

Q.4 Slip in belt drives will:

a) Increase velocity ratio

b) Reduce velocity ratio

c) No effect

d) Increase torque

Q.5 In a flywheel, fluctuation of speed is expressed as:

a) Nmax-Nmin b) Nmin/Nmax

c) (Nmax- Nmin) Nmean d) None

Q.6 A cam converts rotary motion into:

a) Linear motion

b) Oscillating motion

c) Reciprocating or oscillating motion

d) Constant speed motion

Q.7 In dynamic balancing, which condition must be satisfied?

a) Only force balance

b) Only couple balance

c) Both force and couple balance

d) No conditions required

Q.8 In longitudinal vibrations, particles of shaft move:

a) Parallel to axis b) Perpendicular to axis

c) Circular path d) Randomly

Q.9 The Whitworth quick return mechanism is an inversion of:

a) Single slider crank b) Double slider crank

c) Four bar chain d) Crank rocker

Q.10 Crowning of pulleys helps to:

a) Reduce slip b) Keep belt centered

c) Increase torque d) Increase load

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define lower pair in kinematics.

Q.12 State one example of higher pair.

Q.13 Formula for velocity ratio of belt drive without slip.

Q.14 Define slip in belt drive.

Q.15 Write one application of flywheel.

Q.16 Name any two types of cam followers.

Q.17 Define reference plane in balancing.

Q.18 What is torsional vibration?

Q.19 Define coefficient of fluctuation of energy.

Q.20 State one limitation of V-belt drives.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain different types of kinematic pairs with examples.

Q.22 Describe Grashof's law and its importance.

Q.23 Explain double slider crank inversions with neat sketches.

Q.24 Differentiate between open and cross belt drives.

Q.25 Explain the effect of slip and creep in belt drive performance.

Q.26 Explain with a sketch the working of a fast and loose pulley drive.

Q.27 A flywheel has a mass of 250 kg and radius of gyration 0.5 m. It runs at a mean speed of 300 rpm. The fluctuation of energy is 3.6 kj. Calculate the coefficient of fluctuation of speed.